

DiD_Modify_11/08/2025

Difference-in-Differences (DiD) specification

Main findings

Difference-in-Differences (DiD) specification

The study estimate the treatment effect of drought on cropland NDVI and net migration using the canonical two-way fixed effects (TWFE) model:

$$Y_{it} = \alpha + \beta_1 \text{Treated}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Treated}_i \times \text{Post}_t) + \gamma_i + \delta_t + \varepsilon_{it}$$

Where:

- Y_{it} is the outcome for county i in year t ;
- Treated_i is a time-invariant indicator equal to 1 for counties exposed to drought;
- Post_t is an indicator equal to 1 for years after 2010;
- $\text{Treated}_i \times \text{Post}_t$ is the DiD interaction term (β_3), capturing the average treatment effect on the treated (ATT);
- γ_i and δ_t are county and year fixed effects;
- ε_{it} is the error term.

Standard errors are clustered at the county level.

Note: Treated_i and Post_t main effects are absorbed by fixed effects and not separately identified.

Table 5.1 Regression results (DiD main effect)

Variable	NDVI (cropland) Coef (SE)	Net migration per 1000 Coef (SE)
Treated_i	FE absorbed	FE absorbed
Post_t	FE absorbed	FE absorbed
Treated _i × Post _t	-0.0179* (0.0040)	-3.8917 (4.0100)
R-squared	0.976	0.459

Variable	NDVI (cropland) Coef (SE)	Net migration per 1000 Coef (SE)
Adj. R²	0.970	0.313
N	416	416
Significance	*** p<0.01	n.s.

Main findings

- In drought-affected counties, cropland NDVI declined by approximately 0.018 after 2010, representing around 2–3% of the pre-drought mean. This effect is highly statistically significant ($p < 0.01$).
- For net migration, the interaction coefficient is negative (–3.89), indicating greater out-migration from treatment counties post-2010, but the effect is not statistically significant.
- The direction of the migration effect is consistent across specifications, though with wide confidence intervals.
- The fixed effects absorb the main effects of Treated i and Post t , which is expected in a TWFE framework.